A note on Open AI's "A counterexample to Hadwiger's conjecture"

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1 Summary

The manuscript constructs a triangle-free graph G whose complement $H = \bar{G}$ satisfies $\alpha(H) \leq 2$ and $\mathrm{cm}(H) < m/100$, where m is the number of vertices. The connected matching number is defined as follows:

Definition 1. *A matching M in a graph H is called a* connected matching *if every two distinct edges of M are joined by an edge of H: for all distinct $e, f \in M$, there exist $u \in V(e)$ and $v \in V(f)$ such that $uv \in E(H)$. The* connected matching number *is*

$$\mathrm{cm}(H) = \max\{|M| : M \text{ is a connected matching in } H\}.$$

The complement graph provides a counterexample to Hadwiger's conjecture, which asserts that $\chi(H) \leq h(H)$, where $h(H)$ is the largest integer t such that H contains a K_t minor. Indeed, Proposition 3.5 of the paper (proved in Appendix A) gives

$$h(H) \leq \frac{|V(H)| + 4\,\mathrm{cm}(H) + 2}{3}.$$

Applying this to $H = \bar{G}$, and using $\alpha(\bar{G}) \leq 2$ so that every color class has at most two vertices, we obtain, for m large enough,

$$h(\bar{G}) \leq \frac{m + 4\,\mathrm{cm}(\bar{G}) + 2}{3} < \frac{26m}{75} + \frac{2}{3} < \frac{m}{2} \leq \chi(\bar{G}).$$

Thus $h(\bar{G}) < \chi(\bar{G})$, contrary to Hadwiger's conjecture.

2 The construction

2.1 Triangle-free graphs from linear algebra

We first describe the general linear-algebraic scheme and then specify the data used in the manuscript. All vector spaces, ranks, and matrix operations are over $\mathbb{F}_2$.

Let V, W be finite-dimensional vector spaces. Fix $a \in V^*$ and a symmetric bilinear form $T : V \times V \to \mathbb{F}_2$, and put

$$r(\lambda) = a + T(\lambda, \cdot) \in V^*.$$

Let Ω be the set of vertices. For each $o \in \Omega$, assign an injective linear map $U_o : V \to W$ and a linear functional $u_o \in W^*$. Then we can define the associated graph:

Definition 2. *Two vertices $i, j \in \Omega$ are adjacent if there exist $\lambda_i, \lambda_j \in V$ such that*

$$U_i\lambda_i = U_j\lambda_j, \tag{1}$$

$$u_j \circ U_i = r(\lambda_i), \qquad u_i \circ U_j = r(\lambda_j), \tag{2}$$

$$a(\lambda_i) + a(\lambda_j) = 1. \tag{3}$$

Equation (2) *is an equality in V^*.*

Lemma 1. *The associated graph is loopless and triangle-free.*

Proof. Loopless is guaranteed by injectivity of U and (3). Suppose i, j, k is a triangle. Write λ_{ij} for the λ_i at i for the pair ij. Evaluate (2) on λ_{ik}:

$$u_j(U_i\lambda_{ik}) = a(\lambda_{ik}) + T(\lambda_{ij}, \lambda_{ik}).$$

Then sum six equations by permuting i, j, k. For fixed j, the two terms on the left cancel because $U_i\lambda_{ik} = U_k\lambda_{ki}$. For fixed i, the two bilinear terms cancel by symmetry of T. The remaining right-hand side is

$$\sum_{\{i,j\}} \big(a(\lambda_{ij}) + a(\lambda_{ji})\big) = 1 + 1 + 1 = 1,$$

whereas the left-hand side is zero, a contradiction. □

2.2 Construction of U and u

Write $\mathrm{Mat}_{m,n}(\mathbb{F}_2)$ for the space of $m \times n$ binary matrices, and $\mathrm{Mat}_n(\mathbb{F}_2) = \mathrm{Mat}_{n,n}(\mathbb{F}_2)$. Fix a finite index set $\mathcal{E}$, positive integers N, d_B, h with $N \geq 2(d_B + h)$, and a matrix $E \in \mathrm{Mat}_{d_B}(\mathbb{F}_2)$, to be specified later. Take

$$V \subseteq \bigoplus_{e \in \mathcal{E}} \mathrm{Mat}_{d_B}(\mathbb{F}_2) \quad \text{a linear subspace,} \qquad W = \bigoplus_{e \in \mathcal{E}} \mathrm{Mat}_N(\mathbb{F}_2).$$

For $x \in V$, write $x = (x_e)_{e \in \mathcal{E}}$, so $x_e \in \mathrm{Mat}_{d_B}(\mathbb{F}_2)$.

At each component e, consider quadruples

$$P_e, Q_e \in \mathrm{Mat}_{N,d_B}(\mathbb{F}_2), \qquad X_e, Y_e \in \mathrm{Mat}_{N,h}(\mathbb{F}_2)$$

satisfying

$$\begin{aligned}&[P_e\ X_e] \text{ and } [Q_e\ Y_e] \text{ have full column rank,}\\ &P_e^T Q_e = E,\\ &X_e^T Q_e = 0, \qquad P_e^T Y_e = 0.\end{aligned}$$

The condition $N \geq 2(d_B + h)$ ensures there is enough admissible quadruples. Let $\mathcal{O}$ be the set of tuples $o = (P_{o,e}, Q_{o,e}, X_{o,e}, Y_{o,e})_{e\in\mathcal{E}}$ of admissible quadruples. Define a probability law on $\mathcal{O}$ by sampling uniformly at each component, independently across components.

For matrices of the same size, define $\langle A, B\rangle = \mathrm{Tr}(A^T B)$. Each $o \in \mathcal{O}$ determines $U_o : V \to W$ and $u_o \in W^*$ by

$$U_o(x) = (P_{o,e} x_e Q_{o,e}^T)_{e\in\mathcal{E}}, \qquad u_o(\xi) = \sum_{e\in\mathcal{E}} \langle Y_{o,e} X_{o,e}^T, \xi_e\rangle.$$

for $x \in V$ and $\xi \in W$. Each image component $P_{o,e} x_e Q_{o,e}^T$ belongs to $\mathrm{Mat}_N(\mathbb{F}_2)$. Left inverses of $P_{o,e}$ and $Q_{o,e}$ recover x_e from that component, so U_o is injective. Thus these maps provide the vertex data required in Section 2.1; the particular subspace V and the forms a, T will be specified below.

Moreover,

$$u_o(U_o(x)) = \sum_{e\in\mathcal{E}} \mathrm{Tr}\big(X_{o,e} Y_{o,e}^T P_{o,e} x_e Q_{o,e}^T\big) = 0,$$

since $Y_{o,e}^T P_{o,e} = 0$.

2.2.1 A model for uniformly generating quadruples

Set $k = d_B + h$, and combine the columns into

$$A = [P\ X], \qquad B = [Q\ Y] \in \mathrm{Mat}_{N,k}(\mathbb{F}_2).$$

The constraints say that A, B have full column rank and

$$A^T B = H_C, \qquad H_C = \begin{pmatrix} E & 0 \\ 0 & C \end{pmatrix}, \qquad C \in \mathrm{Mat}_h(\mathbb{F}_2).$$

Here $C = X^T Y$ is unrestricted. The letter B in this subsection denotes the combined minus frame, not the coefficient space introduced later.

First fix a full-column-rank A. Choose a right inverse D of A^T and a matrix K whose columns form a basis of $\ker A^T$:

$$D \in \mathrm{Mat}_{N,k}(\mathbb{F}_2), \qquad K \in \mathrm{Mat}_{N,N-k}(\mathbb{F}_2), \qquad A^T D = I_k, \quad A^T K = 0.$$

Both can be obtained by Gaussian elimination, using a fixed rule for each A. Every matrix satisfying the linear constraints has a unique expression

$$\boxed{B = DH_C + KZ}, \qquad C \in \mathrm{Mat}_h(\mathbb{F}_2), \quad Z \in \mathrm{Mat}_{N-k,k}(\mathbb{F}_2).$$

Indeed, $A^T B = H_C$ and the remaining columns lie in $\ker A^T$. Since $[D\ K]$ is invertible,

$$\operatorname{rank} B = k \quad \Longleftrightarrow \quad \operatorname{rank} \begin{pmatrix} H_C \\ Z \end{pmatrix} = k \quad \Longleftrightarrow \quad Z|_{\ker H_C} \text{ is injective.}$$

Thus the linear equations have disappeared, leaving only an injectivity condition on the directions not detected by H_C.

To sample uniformly, use the following procedure:

1. Sample A uniformly among full-column-rank matrices in $\mathrm{Mat}_{N,k}(\mathbb{F}_2)$, and compute D, K.
2. Sample C and Z with independent uniform binary entries. If the stacked matrix above does not have rank k, resample both C and Z, keeping A fixed.
3. Set $B = DH_C + KZ$ and split $A = [P\ X]$ and $B = [Q\ Y]$ at column d_B.

For each A, the correspondence between accepted pairs (C, Z) and admissible minus frames B is bijective. Moreover, the acceptance condition depends only on C, Z, E, not on A, so every full-rank A has the same number of admissible completions. Consequently the resulting quadruple is uniform among all admissible quadruples. Repeat this procedure independently over components.

The rejection step has a uniformly bounded expected cost. For fixed C, let $t = \dim \ker H_C$. The restriction of a uniform Z to this kernel is a uniform $(N-k) \times t$ matrix, so its probability of being injective is

$$\prod_{j=0}^{t-1} (1 - 2^{j-(N-k)}),$$

with the empty product equal to one. Since $t \le k \le N - k$, this probability is at least $1/4$: it is bounded below by $\prod_{r=1}^{\infty}(1 - 2^{-r})$, whose first two factors give $3/8$ and whose remaining product is at least $1 - \sum_{r=3}^{\infty} 2^{-r} = 3/4$. Thus fewer than four trials are needed on average. Importantly, fixing a uniformly sampled C and resampling only Z would generally give a different law, because the acceptance probability depends on $\operatorname{rank} C$.

2.3 Parameters

The only asymptotic parameter is n. The manuscript fixes

$$C_0 = 1000, \qquad g = 10^9 + 1, \qquad D = 4C_0 g, \qquad M = 2^{1000}.$$

The constants b, j_*, J, r_0, M_0 are chosen in the dependency order of Appendix A of the manuscript, before n tends to infinity. Their numerical values are not specified. Set

$$h = 1000r_0, \qquad N = M_0 n, \qquad m = 2^{C_0 g N}.$$

Here M is a density bound used in the distribution theorem, whereas M_0 scales the ambient dimension. Choose M_0 sufficiently large to satisfy the dimension and counting requirements, including $N \geq 2(d_B + h)$ for all sufficiently large n. We take $n \geq 2r_0$ and $j_* \geq 1$.

2.4 Boolean moment spaces

Let

$$I = \mathbb{Z}/g\mathbb{Z}, \quad \mathcal{E} = \binom{I}{2}, \quad I_d = \{d+1, \ldots, d+(g-1)/2\} \subset I.$$

Elements of I are *tags* and elements of $\mathcal{E}$ are *components*. A *Boolean point* consists of a *selector* $s \in \mathbb{F}_2^b$ and *base coordinates* z in g^2+3 blocks of n bits:

$$z = \left((z_{O_{d,t}})_{d,t \in I},\ z_S,\ z_\#,\ z_Z\right) \in (\mathbb{F}_2^n)^{g^2+3}.$$

For each block label $A \in \{O_{d,t} : d, t \in I\} \cup \{S, \#, Z\}$, write

$$z_A = (z_{A,1}, \ldots, z_{A,n}) \in \mathbb{F}_2^n.$$

Thus z has g^2+3 blocks, each containing n bits; $z_{A,k}$ denotes the kth bit in block A. Write

$$p_* = \sum_{j=0}^{j_*} \binom{b}{j}, \qquad B = \mathbb{F}_2^{p_*} \otimes \mathbb{F}_2^{1+(g^2+3)n}, \qquad d_B = \dim B.$$

Let $p(s) \in \mathbb{F}_2^{p_*}$ be the bit vector listing the evaluations at s of all squarefree selector monomials of degree at most j_*, in a fixed order. Define the evaluation vector and its point moment by

$$v(s,z) = p(s) \otimes (1, z), \qquad w(s,z) = v(s,z)v(s,z)^T \in \mathrm{Mat}_{d_B}(\mathbb{F}_2).$$

Identify $B \otimes B$ with $\mathrm{Mat}_{d_B}(\mathbb{F}_2)$ using the fixed coefficient basis. Let $\mathcal{M} \subseteq \mathrm{Mat}_{d_B}(\mathbb{F}_2)$ be the span of all point moments. For $\ell \in I$, let $\mathcal{M}_\ell$ be the span of those point moments whose base values satisfy $O_{d,t} = 0$ whenever $\ell \notin I_d$. Let $\mathcal{M}_*$ be the span obtained by setting all O-blocks to zero.

Define the linear cut-profile map

$$\begin{aligned} f : \bigoplus_{\ell \in I} \mathcal{M}_\ell &\longrightarrow \bigoplus_{e \in \mathcal{E}} \mathcal{M}, \\ (w_\ell)_{\ell \in I} &\longmapsto \left(w_\ell + w_{\ell'}\right)_{\{\ell,\ell'\} \in \mathcal{E}}. \end{aligned}$$

Take the witness space in Sections 2.1 and 2.2 to be $V = \operatorname{im} f$. A tuple $(w_\ell)_{\ell \in I}$ is a representation of $x \in V$ if $f((w_\ell)_{\ell \in I}) = x$. By Lemma 2.3 of the manuscript,

$$\ker f = \{(w_*)_{\ell \in I} : w_* \in \mathcal{M}_*\},$$

that is, two representations of the same x differ by a constant tuple with value in $\mathcal{M}_*$.

2.5 The functional a, the bilinear form T, and the matrix E

The witness space is now fixed as $V = \operatorname{im} f$, and the ambient space remains $W = \bigoplus_{e \in \mathcal{E}} \operatorname{Mat}_N(\mathbb{F}_2)$ from Section 2.2. We next specify $a \in V^*$, a symmetric bilinear form T on V, and the matrix E used to constrain the frames.

Define *linear testers* on moments by selecting ordered matrix entries:

$$\eta_{d,t}(w(s,z)) = \sum_{\rho=1}^{r_0} z_{O_{d,t},2\rho-1} z_{O_{d,t},2\rho},\ d,t \in I$$
$$\eta_S(w(s,z)) = \sum_{\rho=1}^{r_0} z_{S,2\rho-1} z_{S,2\rho}.$$

Put $\eta_O = \sum_{d,t} \eta_{d,t}$ and $\eta = \eta_O + \eta_S$. These are elements of $\mathcal{M}^*$ and take values in $\mathbb{F}_2$.

For a cut profile x represented by (w_ℓ), set

$$a_t(x) = \sum_{\ell,d} \eta_{d,t}(w_\ell), \qquad a(x) = \sum_t a_t(x), \qquad b_t(x) = \sum_{\ell \neq t} \eta_S(w_\ell).$$

These functionals do not depend on the representation: the O-testers vanish on $\mathcal{M}_*$, and a constant change contributes $(g-1)\eta_S(w_*) = 0$ to b_t.

Define

$$T_0(x,y) = a(x)a(y) + \sum_{t \in I} (a_t(x) b_t(y) + b_t(x) a_t(y)).$$

For each $1 \le j \le J$ and ordered pair $e, f \in \mathcal{E}$, fix endomorphisms L_j^{ef}, R_j^{ef} of B, chosen to satisfy the uniform mixing conditions of Lemma 5.5 of the manuscript. With the entrywise pairing $\langle A, B \rangle = \operatorname{Tr}(A^T B)$, put

$$B_1(x,y) = \sum_{j=1}^{J} \sum_{e,f \in \mathcal{E}} \left\langle x_e, L_j^{ef} y_f (R_j^{ef})^T \right\rangle,$$
$$T_1(x,y) = B_1(x,y) + B_1(y,x), \qquad T = T_0 + T_1.$$

Then T is symmetric and $T(x,x) = a(x)$. This specifies $r(\lambda) = a + T(\lambda, \cdot)$ in the edge definition.

The matrix $E \in \mathrm{Mat}_{d_B}(\mathbb{F}_2)$ prescribes $P_{o,e}^T Q_{o,e} = E$ in Section 2.2. Let E_η be the sum of the ordered tester matrices, so $\langle E_\eta, w\rangle = \eta(w)$ for $w \in \mathcal{M}$. Choose $M_1, \dots, M_b \in \mathrm{Mat}_n(\mathbb{F}_2)$ along with the mixers as in Lemma 5.5, and define $E_\#$ by

$$v(s,z)^T E_\# v(s',z') = \sum_{\rho=1}^{b} (s_\rho + s'_\rho) z_\#^T M_\rho z'_\#.$$

This is bilinear on B since $j_* \geq 1$. Set $E = E_\eta + E_\#$. Its additional term vanishes on coincident point inputs, so

$$\langle E, w\rangle = \eta(w) \qquad (w \in \mathcal{M}).$$

The Z-block does not occur in E. The mixers and the matrices M_ρ are chosen once for each sufficiently large n, before graph sampling. Their existence with the required rank properties is supplied by Lemma 5.5; arbitrary choices need not give the connected-matching bound.

2.6 The vertex law

With $\mathcal{E}, d_B, N, h, V, a, T, E$ now specified, apply Section 2.2. Let $\mathcal{O}_n$ be the set of admissible frame tuples and let μ_n be its product uniform law across components. An element $o \in \mathcal{O}_n$ determines the maps $U_o : V \to W$ and $u_o \in W^*$ already defined there. In the general scheme of Section 2.1, take $\Omega = \mathcal{O}_n$.

The initial random variable is therefore a tuple of frame matrices. The coefficient space, moment spaces, a, T, and E are fixed before drawing these tuples. In particular, the matrix E alone does not specify the graph: adjacency also depends on V, a, T.

2.7 The sampled graph and the remaining theorem

Sample

$$o_1, \dots, o_m \overset{\text{iid}}{\sim} \mu_n, \qquad m = 2^{C_0 g N}.$$

On the positions $1, \dots, m$, put $ij \in E(G)$ exactly when o_i, o_j satisfy the edge criterion (1)–(3); put $ij \in E(H)$ otherwise. This completes the definition of the random graph. Its edges are determined by the sampled frames, with no further independent edge sampling.

Remark 1. *The initial random variable is the tuple of frame matrices o, not a Boolean point (s,z). The monomial list and moment spaces are fixed globally. Boolean points vary when constructing witnesses; sampling point parameters with the frames fixed is called a query in the manuscript.*

Triangle-freeness of G, and hence $\alpha(H) \leq 2$, holds for every sample. The harder assertion is that $\mathrm{cm}(H) < m/100$ with probability $1 - \exp(-\Omega(m))$ (Proposition 3.4). To obtain it,

the manuscript calls an ordered pair of raw vertices that do not satisfy the edge criterion a *unit*; such a pair represents an edge of H when placed at distinct sampled positions. Two units conflict if all four cross pairs are edges of G, so no edge of H joins their endpoint sets. Their corresponding edges therefore cannot coexist in a connected matching.

Theorem 3.1 asserts that every law σ on units satisfying the pointwise density bounds

$$\sigma_1 \le M\mu_n, \qquad \sigma_2 \le M\mu_n, \qquad \sigma \le 2^{DN}(\mu_n \otimes \mu_n)$$

gives a conflict between two independent σ-units with probability at least 2^{-100gN}. The endpoints within one unit may be dependent. An entropy-based container argument transfers this statement to the sampled graph. Thus the construction above specifies the graph; the subsequent distribution theorem and container argument establish the small connected matching number needed for the counterexample.

3 From Theorem 3.1 to the counterexample

Assume Theorem 3.1 of the manuscript. We explain how it implies that the sampled graph $H = \bar{G}$ has $\mathrm{cm}(H) < m/100$ with probability $1 - \exp(-\Omega(m))$. Throughout this section write $\Omega = \mathcal{O}_n$, $\mu = \mu_n$, and $\mu^2 = \mu \otimes \mu$.

3.1 The distribution theorem and the conflict graph

Let $\mathcal{U} \subseteq \Omega^2$ be the set of units: ordered pairs that do not satisfy the edge criterion for G. Define a conflict graph Γ on $\mathcal{U}$, joining (a_1, a_2) to (b_1, b_2) when all four cross pairs $a_i b_j$ satisfy that criterion. This graph has no loops, since a unit cannot conflict with itself. A connected matching of H, after orienting its edges, gives an independent set of unit types in Γ. Different matching edges may have the same type; we keep track of their occurrences in the sample separately.

For a probability law σ on units, its marginals are

$$\sigma_1(a) = \sum_b \sigma(a, b), \qquad \sigma_2(b) = \sum_a \sigma(a, b).$$

Call σ feasible if the following inequalities hold pointwise:

$$\sigma_1 \le M\mu, \qquad \sigma_2 \le M\mu, \qquad \sigma \le 2^{DN}\mu^2. \tag{4}$$

These are upper bounds on concentration, rather than distances between probability laws. Theorem 3.1 says that two independent draws from any feasible law conflict with probability at least

$$\varepsilon = 2^{-100gN}.$$

The law need not be uniform, and its two endpoints need not be independent.

3.2 Containers obtained by deleting conflicts

Fix an independent set $I \subseteq \mathcal{U}$. Start with the residual family $R = \mathcal{U}$. Whenever R supports a feasible law, choose one, denoted ρ_R, minimizing relative entropy with respect to μ^2:

$$\mathcal{D}(\rho_R \| \mu^2) = \sum_{a,b} \rho_R(a,b) \log \frac{\rho_R(a,b)}{\mu(a)\mu(b)}.$$

Here logarithms are natural and zero summands are omitted. By Theorem 3.1,

$$\sum_{v \in R} \rho_R(v) \rho_R(N_R(v)) \geq \varepsilon,$$

where $N_R(v)$ is the conflict neighborhood of v in R. Choose v maximizing $\rho_R(N_R(v))$, so this mass is at least ε. Make all choices deterministic, using a fixed ordering to break ties.

If $v \in I$, record v and delete $N_R(v)$; if $v \notin I$, delete v alone. Both steps preserve $I \subseteq R$. Stop when no feasible law remains. The recorded units form a fingerprint, and the terminal family R is its container.

The entropy bound limits the number of recordings. Every feasible law has

$$0 \leq \mathcal{D}(\rho \| \mu^2) \leq DN \log 2.$$

The information-projection inequality of Lemma 3.2 says that, after a recording which leaves a feasible residual family, the minimum entropy increases by at least $-\log(1-\varepsilon)$. Indeed, the old minimizer assigns mass at most $1-\varepsilon$ to the new residual family, and the new law is supported there. Other deletions cannot decrease the minimum entropy. Allowing one final recording that destroys feasibility, the number of recordings is at most

$$L_N = 1 + \left\lceil \frac{DN \log 2}{-\log(1-\varepsilon)} \right\rceil = O(N 2^{100gN}).$$

The fingerprint determines the terminal family by replaying the deterministic procedure: a selected unit is recorded exactly when it belongs to the fingerprint. A previously recorded unit has no remaining conflict neighbors and cannot be selected again while feasibility holds.

There are at most $2^{2|\mathcal{E}|N(d_B+h)}$ raw vertices, so $\log|\mathcal{U}| = O(N^2)$. Hence the number of containers is at most $(|\mathcal{U}|+1)^{L_N}$, whose logarithm is

$$O(N^3 2^{100gN}) = o(m), \qquad m = 2^{1000gN}.$$

Thus every independent set lies in one of $\exp(o(m))$ terminal containers, all determined before the graph sample is drawn.

3.3 A terminal container has rare endpoints or rare pairs

If a family R supports no feasible law, Lemma 3.3 gives sets $S \subseteq \Omega$ and $E_0 \subseteq \Omega^2$ such that

$$\mu(S) < 1/M, \qquad \mu^2(E_0) < 2^{-DN},$$

and each $(a, b) \in R$ has $a \in S$, $b \in S$, or $(a, b) \in E_0$.

To see why, put a left and a right copy of Ω in a flow network. Give the edges from the source to left a capacity $M\mu(a)$, those from right b to the sink capacity $M\mu(b)$, and the edge from left a to right b capacity $2^{DN}\mu(a)\mu(b)$ whenever $(a, b) \in R$. A flow of value one is precisely a feasible law. If none exists, a cut of capacity less than one supplies sets $L_0, R_0 \subseteq \Omega$ with

$$M\mu(L_0) + M\mu(R_0) + 2^{DN}\mu^2\big(R \cap ((\Omega \setminus L_0) \times (\Omega \setminus R_0))\big) < 1.$$

Take $S = L_0 \cup R_0$ and let E_0 be the pair set in the last term. This proves the asserted cover and measure bounds.

3.4 The random sample excludes large connected matchings

Fix one terminal container and its sets S, E_0, and put $k = \lceil m/200 \rceil$. For independent sampled positions $o_1, \dots, o_m$, the probability that at least k lie in S is at most

$$\binom{m}{k}\mu(S)^k \le \left(\frac{\mathrm{e}m}{kM}\right)^k \le (200\mathrm{e}/M)^k = \exp(-\Omega(m)).$$

Here e is Euler's number. A fixed collection of k disjoint ordered position pairs consists of independent μ^2 draws. The probability they all belong to E_0 is at most 2^{-DNk}. There are at most m^{2k} such collections, so

$$\Pr(k \text{ disjoint sampled pairs lie in } E_0) \le m^{2k}2^{-DNk} = 2^{(2C_0g-D)Nk} = 2^{-2C_0gNk}.$$

Taking a union bound over the $\exp(o(m))$ containers, with probability $1 - \exp(-\Omega(m))$ every container has fewer than $m/200$ sampled positions in S and fewer than $m/200$ disjoint sampled pairs in E_0.

Suppose on this event that H has a connected matching of size at least $m/100$. Orient its edges. Their unit types form an independent set in Γ, so they belong to a terminal container. Because the matching uses disjoint positions, fewer than $m/200$ of its edges can meet that container's sampled positions in S. All remaining edges belong to E_0, and there are fewer than $m/200$ of those as well. This contradicts the assumed matching size. Therefore

$$\mathrm{cm}(H) < m/100.$$

No assumption that the uniform law on the matching itself is feasible is needed: the container procedure applies to its independent set of types regardless.

Finally, $\alpha(H) \le 2$ holds for every sample. Applying Appendix A, for $m \ge 5$ we obtain

$$h(H) \le \frac{m + 4\,\mathrm{cm}(H) + 2}{3} < \frac{26m}{75} + \frac{2}{3} < \frac{m}{2} \le \chi(H).$$

Thus Theorem 3.1 implies that the construction produces a counterexample with probability $1 - \exp(-\Omega(m))$ for all sufficiently large n.

A The clique-minor bound (Proposition 3.5)

Proposition 1. *For every finite nonempty graph H with m vertices,*

$$h(H) \le \frac{m + 4\,\mathrm{cm}(H) + 2}{3}.$$

Proof. Write $c = \mathrm{cm}(H)$ and $b = h(H)$. A K_b minor model consists of b disjoint nonempty connected branch sets, with an edge joining every pair. Let s be the number of singleton branch sets and e the number of two-vertex branch sets.

Each two-vertex branch set supplies an edge. The singleton vertices form a clique, so pairing them supplies $\lfloor s/2 \rfloor$ more edges. Together these form a connected matching, since every pair of its edges is joined by an edge of H. Thus

$$c \ge e + \left\lfloor \frac{s}{2} \right\rfloor, \qquad s \le 2(c - e) + 1.$$

Every remaining branch set has at least three vertices, so

$$m \ge s + 2e + 3(b - s - e) = 3b - 2s - e.$$

Consequently,

$$3b \le m + 2s + e \le m + 4c - 3e + 2 \le m + 4c + 2,$$

which proves the claim. □